

# Balanced orbit defects in half-turn-symmetric no-three-in-line configurations, with new $2n$ -point configurations for $n = 36, 37, 39$

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## Abstract

A  $2n$ -point no-three-in-line configuration on the  $n \times n$  grid with half-turn symmetry can be written as the union of the orbits of a larger subgroup  $H \leq D_4$  (the quarter turns, or the two diagonal reflections) that it contains, together with a set of half-turn pairs that are not  $H$ -orbits — its *orbit defect* relative to  $H$ . We prove a *balance lemma*: the defect pairs, read as arcs on the row classes  $\{i, n-1-i\}$ , form a balanced directed multigraph, hence a union of directed cycles. With one further observation (two half-turn pairs on the same long diagonal are collinear) this classifies the defects with at most three pairs: one pair must be a diagonal loop — Flammenkamp’s pseudo-class rct4; two pairs are two loops on different diagonals or a directed 2-cycle, which is an orbit of the diagonal reflections; three pairs form a directed 3-cycle or a loop plus a 2-cycle. We enumerate the corresponding families exhaustively for small  $n$  with an exact branch-and-bound program (validated against OEIS A000769 and the class counts of Flammenkamp’s database): the 3-cycle family for odd  $n \leq 25, 33$  and (in part)  $37, 39, 41$ ; the two-loop family for even  $n \leq 28$  and  $n = 36$ ; the mixed families for  $n \leq 27$  (all empty). Among the configurations found are: for  $n = 36$ , the first located  $2n$ -point configurations whose symmetry group is exactly the half-turn (exactly three inequivalent ones in the two-loop family); for  $n = 37$  (two) and  $n = 39$  (three), the first located configurations of that symmetry group outside the rct4 pseudo-class (Flammenkamp’s rct4 configurations of 1997/98 already have exactly half-turn symmetry); and a further one for  $n = 33$ . “First located” refers to the public files of Flammenkamp’s database as of 2026-08-11 and awaits confirmation by its maintainer. Configurations, programs and sweep journals are public.

## 1 Introduction

The no-three-in-line problem (Dudeney, 1917) asks for the largest set of points of the  $n \times n$  grid  $\{0, \dots, n-1\}^2$  containing no three collinear points. Since a line parallel to an axis contains at most two points, at most  $2n$  points can be placed, and  $2n$ -point configurations are the objects of interest. Their existence for every  $n$  is a well-known open problem; a probabilistic heuristic (Guy–Kelly, corrected form) suggests that for large  $n$  fewer than  $2n$  points can be placed. Computationally,  $2n$ -point configurations were known for all  $n \leq 46$  and for  $n = 48, 50, 52$  for decades [1, 2]; in 2025–2026 constraint programming [4], GPU branch-and-bound [5] and SAT solving extended this to all  $n \leq 70$  and to  $n = 72, 74, 76$  (see the database [3], which also records, for each  $n$ , the number of configurations by their exact symmetry class under the dihedral group  $D_4$  of the square, and marks the cells for which no configuration is known).

This note concerns those cells rather than the record sizes. Flammenkamp’s table lists, for each  $n$ , the number of inequivalent  $2n$ -point configurations whose stabilizer in  $D_4$  is exactly a given subgroup: trivial (*iden*), the half-turn (*rot2*), one diagonal reflection (*dia1*), one axis

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reflection (*ort1*), the quarter turns (*rot4*), both diagonal reflections (*dia2*), both axis reflections (*ort2*) and the full group (*full*). Inside the half-turn class he keeps, for odd  $n$ , the pseudo-class *rct4* (“rot4 symmetry except on the long diagonals”: quarter-turn orbits off the diagonals plus one half-turn pair on one diagonal), whose configurations have stabilizer exactly the half-turn but are listed in a separate column *c*, the column *:* containing the remaining rot2 configurations [3]. The *rct4* configurations were found by Flammenkamp around the turn of 1996/97 and published in [2]; the odd- $n$  construction of [4] is of the same type. As of 2026-08-11 the column *:* was complete for  $n \leq 31$ ; for  $n = 36$  both columns were empty, and for  $n = 37, 39$  the column *:* was empty while column *c* listed 21 and 33 configurations.

**Results.** (i) A balance lemma (Section 2) describing which orbit defects a half-turn-symmetric configuration may have relative to a larger subgroup  $H \leq D_4$ : the defect pairs must form a balanced directed multigraph on the row classes. Together with the collinearity of two pairs on one diagonal this classifies the defects with at most three pairs (Proposition 3): a single pair is a diagonal loop (this is exactly the *rct4* structure); two pairs are two loops on different diagonals or a directed 2-cycle, and a directed 2-cycle is itself an orbit of the diagonal reflections; three pairs form a directed 3-cycle or a loop plus a 2-cycle. (ii) Exhaustive enumerations of the resulting families for small  $n$  (Section 4), with all sub-classes listed and their journals kept: the 3-cycle family, the two-loop family and the mixed families. (iii) New  $2n$ -point configurations with stabilizer exactly the half-turn: three inequivalent ones for  $n = 36$  (two-loop family; both database cells were empty), two for  $n = 37$  and three for  $n = 39$  (3-cycle family; the first located configurations of that stabilizer that are not of *rct4* type), and one more for  $n = 33$ ; the configurations are listed in the Appendix and were verified by independent programs. (iv) A description of the exact solver and its validation, measured growth of the cost of finding one  $2n$ -point configuration in the classes *rot4*, *rct4* and *rot2*, and a list of negative results with their exact scope.

**Related work.** The best known lower bound,  $3n/2 - o(n)$  points, is the modular-hyperbola construction of Hall, Jackson, Sudbery and Wild [7]; it has not been improved since 1975 (see [8, 9] and Problem 72 of [11]). For  $k \geq 3$  points per line the asymptotic answer  $kn$  was obtained recently by probabilistic methods [8, 9], which explicitly do not apply to  $k = 2$ . The probabilistic heuristic of Guy and Kelly [6], in its corrected form [10, 4], predicts that  $2n$ -point configurations become rare for large  $n$ ; Prellberg notes that the same heuristic wrongly predicts fewer than one symmetric configuration [4], and Flammenkamp explains the abundance of rotationally symmetric configurations by the fact that a pair of points blocks only  $O(\log n)$  cells in the rotational classes against  $\Theta(n)$  in the axis-reflection classes [3]. Machine-learning approaches (PatternBoost, reinforcement learning) have been compared with integer programming on this problem and do not scale beyond  $n \approx 14$  [12]. The abstract content of the balance lemma — a zero-divergence condition and its cycle decomposition — is standard (see e.g. [13] for balanced two-coloured degree sequences and alternating cycles); its application to symmetric no-three-in-line configurations, and the resulting search families, we have not found in the literature. Flammenkamp uses “defect” for the shortfall from  $2n$  points; to avoid a clash we say *orbit defect*.

## 2 Symmetry classes and the balance lemma

Let  $m = n - 1$  and let  $D_4$  act on the grid; write  $\rho(u, v) = (v, m - u)$  for the quarter turn, so  $\rho^2(u, v) = (m - u, m - v)$ ,  $\sigma(u, v) = (v, u)$  for the reflection in the main diagonal and  $\sigma'(u, v) = (m - v, m - u) = \rho^2\sigma(u, v)$  for the reflection in the anti-diagonal. A  $2n$ -point no-three-in-line configuration has exactly two points in every row and every column.

Fix a subgroup  $H \leq D_4$  containing  $\rho^2$  and a configuration  $C$  with  $\rho^2 C = C$ . Let  $B$  be the union of all  $H$ -orbits contained in  $C$  (the  $H$ -base of  $C$ ; it is  $H$ -invariant, and it is determined

by  $C$  and  $H$ ). Since  $C$  is  $\rho^2$ -invariant and  $\rho^2 \in H$ , the complement  $D = C \setminus B$  is a union of half-turn pairs  $\{p, \rho^2 p\}$ , none of which is an  $H$ -orbit;  $D$  is the *orbit defect* of  $C$  relative to  $H$ . For  $a \in \{0, \dots, m\}$  let  $r_X(a)$ ,  $c_X(a)$  denote the numbers of points of a set  $X$  in row  $a$  and in column  $a$ .

**Lemma 1** (balance). *Suppose that  $H$  contains a motion  $g$  with  $g(\{\text{column } a\}) = \{\text{row } a\}$  for every  $a$  (the quarter turn  $\rho$  and the reflection  $\sigma$  have this property; note that  $\sigma'$  alone maps column  $a$  onto row  $m - a$ , but  $\sigma' \in H$  implies  $\sigma = \rho^2 \sigma' \in H$ ). Then every  $H$ -orbit  $O$  satisfies  $r_O(a) = c_O(a)$  for all  $a$ , and consequently the orbit defect  $D$  of a  $2n$ -point configuration satisfies  $r_D(a) = c_D(a)$  for all  $a$ .*

*Proof.*  $g$  maps  $O \cap \{\text{column } a\}$  bijectively onto  $O \cap \{\text{row } a\}$  because  $gO = O$ ; hence  $r_O(a) = c_O(a)$ . Summing over the orbits in  $B$  and using  $r_C(a) = c_C(a) = 2$  gives  $r_D(a) = c_D(a)$ .  $\square$

Interpret the pairs of  $D$  as arcs of a directed multigraph on the *row classes*  $\bar{x} = \{x, m - x\}$  (for odd  $n$  the central class  $\{m/2\}$  is a singleton): the pair  $\{(x, y), (m - x, m - y)\}$  occupies rows  $x, m - x$  and columns  $y, m - y$ , i.e. it is an arc from  $\bar{x}$  to  $\bar{y}$  (a loop if  $\bar{x} = \bar{y}$ ; for  $\bar{x} \neq \bar{y}$  two different pairs can join the same two classes). A pair with  $x = m/2$  has both points in the central row and is an arc out of the central class; a pair with  $y = m/2$  is an arc into it; the central cell  $(m/2, m/2)$  cannot belong to a  $\rho^2$ -invariant configuration with two points per row, but the central class may well be used. Then  $r_D(\bar{x}) = 2 \cdot \text{outdeg}(\bar{x})$  and  $c_D(\bar{x}) = 2 \cdot \text{indeg}(\bar{x})$ , and Lemma 1 says:

**Corollary 2.** *The orbit defect  $D$  is a balanced directed multigraph on the row classes (in-degree = out-degree at every class), hence its arc set decomposes into directed cycles (loops and 2-cycles allowed). In particular a single defect pair must be a loop, i.e. must lie on a long diagonal. For  $H = \langle \rho \rangle$  every  $H$ -orbit in  $C$  has four points (the centre cell cannot occur), so  $2n = 4a + 2|D|$  and  $|D| \equiv n \pmod{2}$ ; for  $H = \langle \sigma, \sigma' \rangle$  the diagonal 2-orbits  $\{(i, i), (m - i, m - i)\}$ ,  $\{(i, m - i), (m - i, i)\}$  are  $H$ -orbits, and if  $b$  of them occur then  $b + |D| \equiv n \pmod{2}$ .*

For  $H = \langle \rho \rangle$  (rot4) an  $H$ -orbit off the diagonals is  $\{(u, v), (v, m - u), (m - u, m - v), (m - v, u)\}$ ; for  $H = \langle \sigma, \sigma' \rangle$  (dia2) it is  $\{(u, v), (v, u), (m - u, m - v), (m - v, m - u)\}$ . We call these the C4 base and the V2 base. The following proposition lists all orbit defects with at most three pairs; the families it names are then the natural next steps beyond rct4.

**Proposition 3** (orbit defects with at most three pairs). *Let  $C$  be a  $2n$ -point configuration with  $\rho^2 C = C$  and let  $D$  be its orbit defect relative to  $H = \langle \rho \rangle$ .*

- (a)  $|D| = 1$ :  $D$  is one loop, i.e. one half-turn pair on a long diagonal ( $n$  odd) — the rct4 structure.
- (b)  $|D| = 2$  ( $n$  even): either two loops at different classes on different diagonals,  $\{(x, x), (m - x, m - x)\}$  and  $\{(y, m - y), (m - y, y)\}$  with  $\bar{y} \neq \bar{x}$  (**two-loop family**); or a directed 2-cycle  $\bar{x} \rightarrow \bar{y} \rightarrow \bar{x}$ , which is a dia2-orbit  $\{(u, v), (v, u), (m - u, m - v), (m - v, m - u)\}$  with  $\bar{u} = \bar{x}$ ,  $\bar{v} = \bar{y}$  that is not a C4-orbit (**mixed family C4 base + one V2 orbit**).
- (c)  $|D| = 3$  ( $n$  odd): either a directed 3-cycle on three distinct classes,  $(x, y), (y', z), (z', x')$  with  $\bar{y}' = \bar{y}$ ,  $\bar{z}' = \bar{z}$ ,  $\bar{x}' = \bar{x}$ , one of the classes possibly the central one (**3-cycle family**); or one loop plus a directed 2-cycle, i.e. a diagonal pair plus a dia2-orbit that is not a C4-orbit (**mixed family with a loop**).

The same list, with the roles of C4- and dia2-orbits exchanged, holds for  $H = \langle \sigma, \sigma' \rangle$ ; there a directed 2-cycle is a C4-orbit that is not a dia2-orbit (**mixed family V2 base + one C4 orbit**), and a loop is a diagonal 2-orbit and hence never a defect.

*Proof.* By Corollary 2,  $D$  is a union of directed cycles on the classes. Two loops on the same diagonal give four collinear points, and two loops at the same class  $\bar{x}$  on the two diagonals form the C4-orbit of  $(x, x)$ , which belongs to the base by definition; this leaves (a) and the first case of (b), and excludes three loops. A directed 2-cycle  $\bar{x} \rightarrow \bar{y} \rightarrow \bar{x}$  consists of two pairs  $\{(u, v), (m - u, m - v)\}$  and  $\{(v', u'), (m - v', m - u')\}$  with  $\bar{u} = \bar{u}' = \bar{x}$ ,  $\bar{v} = \bar{v}' = \bar{y}$ ; the second pair is the image of the first under  $\sigma$  (if  $(v', u') \in \{(v, u), (m - v, m - u)\}$ ) or under  $\rho$  (if  $(v', u') \in \{(v, m - u), (m - v, u)\}$ ); in the second case the union is the C4-orbit of  $(u, v)$  and belongs to the base, in the first case it is the dia2-orbit of  $(u, v)$ . For  $|D| = 3$  the balanced multigraphs are a 3-cycle, a loop plus a 2-cycle, or three loops. For the V2 base the argument is the same with  $\rho$  and  $\sigma$  exchanged.  $\square$

A configuration of one of these families is not  $H$ -symmetric (the defect breaks  $H$ ); its stabilizer in  $D_4$  is some subgroup containing  $\rho^2$  and not containing  $H$ , and it must be computed: for every configuration reported below the stabilizer was determined by testing all eight motions, and it is exactly  $\{1, \rho^2\}$  in every case, so the configurations fall into the rot2 column of the database. (For  $n$  even, a two-loop configuration is “rot4 symmetric except on the long diagonals”; the database keeps rct4 as a pseudo-class for odd  $n$  only, so these configurations belong to the column :.)

### 3 The solver

All searches were done with an exact branch-and-bound program (`there.c`, about 600 lines of C, no external libraries; the family sweeps used the variant `there_twisted_experiment.c` with the two bases and the defect built in). Cells are  $t = un + v$ ; a state keeps two bit-matrices, *alive* by rows and by columns. Every pair of chosen points “shades” all grid cells of the line through them (walked with the primitive direction obtained from a gcd table); shaded cells die, as do all cells of a row or column that already holds two points. A symmetry class is imposed by choosing whole orbits: a cell is a candidate only if its whole orbit is alive, and the “orbit-alive” bit-masks are derived from the alive matrices by transposition and bit reversal (one word operation per group element). Branching uses the most constrained reading class (row or column with the fewest orbit-alive cells) and dies as soon as some class cannot be completed. A failed candidate is killed for its siblings, so that a search with a fixed symmetry class is complete: “no configuration” is a computer-assisted proof of absence for that class (trusted program, no independent certificate). With restarts and randomized tie-breaking the same program is used to find one configuration quickly. A PAIRS option fixes a set of half-turn pairs (the orbit defect), kills their images under the base group, and searches the remaining orbits of the base; the same option with a diagonal pair reproduces rct4 exactly. Invalid PAIRS/FIX input is rejected.

**Validation.** For  $n \leq 10$  the program enumerates all  $2n$ -point configurations without symmetry: 1, 2, 11, 32, 50, 132, 380, 368, 1135 labelled configurations for  $n = 2, \dots, 10$ , which reduce to 1, 1, 4, 5, 11, 22, 57, 51, 156 classes under  $D_4$ , in agreement with OEIS A000769. For symmetric classes we compared the number  $N_H$  of labelled configurations invariant under a fixed subgroup  $H$  with the class counts  $c_K$  of [3]. For a normal subgroup  $H$  (rot2, rot4, dia2, ort2, full) all  $|D_4|/|K|$  labelled images of a configuration of class  $K \supseteq H$  are  $H$ -invariant, so  $N_H = \sum_{K \supseteq H} c_K |D_4|/|K|$ : rot2 for  $n = 8$ :  $36 = 4 \cdot 7 + 2 \cdot 4$ ;  $n = 10$ :  $67 = 4 \cdot 13 + 2 \cdot 6 + 2 \cdot 1 + 1$ ; dia2 for  $n = 11, \dots, 25$ :  $0, 2, 2, 0, 0, 0, 2, 4 = 2 \cdot (0, 1, 1, 0, 0, 0, 1, 2)$ . For  $H = \langle \sigma \rangle$  (dia1) only two of the four images of a dia1 configuration have stabilizer  $\langle \sigma \rangle$  (the other two have  $\langle \sigma' \rangle$ ), so  $N_H = 2c_{\text{dia1}} + 2c_{\text{dia2}} + c_{\text{full}}$ :  $n = 13$ :  $20 = 2 \cdot 9 + 2 \cdot 1$ ;  $n = 15$ :  $28 = 2 \cdot 13 + 2 \cdot 1$ . For rct4 the program’s mode places the diagonal pair on the main diagonal and therefore sees two of the four labelled images of each rct4 configuration:  $n = 9, 17, 19, 21$ :  $2, 2, 4, 2 = 2 \cdot (1, 1, 2, 1)$  — all in agreement. Two independent implementations (the two agents’ programs) agree on every enumeration they both performed.

Every configuration reported here was finally re-checked by an independent script that tests all  $\binom{2n}{3}$  triples in exact integer arithmetic and computes the stabilizer explicitly.

## 4 Results

### 4.1 Exhaustive enumerations of the defect families

The defect sets were enumerated up to  $D_4$  (one representative per orbit of  $D_4$  acting on defects, computed as canonical forms of the six or four cells); for each representative the sub-class (defect + base) was searched exhaustively. Table 1 gives, for odd  $n$ , the number of inequivalent 3-cycle defects —  $4\binom{h}{3} + \binom{h}{2}$  with  $h = (n-1)/2$ , the second term being the cycles through the central class — and the number of pairwise inequivalent configurations found for the two bases; Table 2 gives the two-loop family for even  $n$ ; the mixed families are reported in the text. Rows marked “complete” were swept entirely, every sub-class terminating with “exhausted” (or “dead”, when the fixed defect already violates the law: e.g. 24 of the 2240 non-central sub-classes at  $n = 33$ ); counts marked with  $\geq$  come from sweeps that were still running when this draft was written, with the number of sub-classes already swept stated. *Correction to an earlier draft:* the generator of the 3-cycle lists used until v0.6 omitted the cycles through the central class (an external audit pointed this out); the omitted sub-classes were swept afterwards and produced two additional configurations at  $n = 13$  (V2 base) and  $n = 21$  (C4 base) — both already in the database, whose rot2 column is complete for  $n \leq 31$  — none at  $n = 15, \dots, 25, 33$  and  $37$ , and two new ones at  $n = 39$ ; the corrected counts are given below, and all journals are in the repository (logs/).

| $n$ | defects mod $D_4$ (central) | C4 base  | V2 base  | status of the sweep                                                                                                                                                                                         |
|-----|-----------------------------|----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | 95 (15)                     | 5        | 3        | complete                                                                                                                                                                                                    |
| 15  | 161 (21)                    | 0        | 0        | complete                                                                                                                                                                                                    |
| 17  | 252 (28)                    | 0        | 1        | complete                                                                                                                                                                                                    |
| 19  | 372 (36)                    | 0        | 0        | complete                                                                                                                                                                                                    |
| 21  | 525 (45)                    | 1        | 0        | complete                                                                                                                                                                                                    |
| 23  | 715 (55)                    | 0        | 0        | complete                                                                                                                                                                                                    |
| 25  | 946 (66)                    | 0        | 0        | complete                                                                                                                                                                                                    |
| 33  | 2360 (120)                  | 1        | 0        | complete (both bases)                                                                                                                                                                                       |
| 37  | 3417 (153)                  | 2        | 0        | complete (both bases)                                                                                                                                                                                       |
| 39  | 4047 (171)                  | $\geq 3$ | $\geq 0$ | C4 base: all 171 central sub-classes complete (2 configurations) and 300 of the 3876 non-central swept (1 configuration), rest in progress; V2 base: central complete (none), 3277 non-central swept (none) |
| 41  | 4750 (190)                  | $\geq 0$ | 0        | V2 base complete (none); C4 base: all 190 central sub-classes complete (none), 3030 of the 4560 non-central swept (none), rest in progress                                                                  |
| 45  | 6391 (231)                  | —        | $\geq 0$ | V2 base: 400 sub-classes swept (none); C4 base not started                                                                                                                                                  |

Table 1: The 3-cycle family (odd  $n$ ): C4- or V2-orbits plus a directed 3-cycle of half-turn pairs; numbers of pairwise inequivalent configurations. A sub-class is one defect (one directed 3-cycle up to  $D_4$ ) together with one base; “complete” means every sub-class was searched exhaustively.

**Mixed families.** The mixed families of Proposition 3 — C4 base plus one dia2-orbit that is not a C4-orbit (plus one diagonal loop for odd  $n$ ), and V2 base plus one C4-orbit that is not a dia2-orbit — were swept exhaustively for all  $9 \leq n \leq 27$  (both parities; e.g. 1210 and 55

| $n$   | sub-classes $\{x, y\} \bmod D_4$ | inequivalent configurations                                                                                                                                                                                                     |
|-------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14–22 | 21, 28, 36, 45, 55               | 0 (complete)                                                                                                                                                                                                                    |
| 24    | 66                               | 2 (complete)                                                                                                                                                                                                                    |
| 26    | 78                               | 1 (complete)                                                                                                                                                                                                                    |
| 28    | 91                               | 1 (complete)                                                                                                                                                                                                                    |
| 36    | 153                              | 3 (complete: three sub-classes contain a configuration, two labelled solutions each, i.e. one configuration up to the diagonal reflections; the sub-class $\{1, 4\}$ needed a rerun without time limit, $4.9 \cdot 10^7$ nodes) |

Table 2: The two-loop family (even  $n$ ): C4-orbits plus one half-turn loop on each long diagonal. A sub-class is an unordered pair of row classes  $\{x, y\}$ ,  $0 \leq x < y < n/2$  ( $x$  = class of the main-diagonal loop,  $y$  = class of the anti-diagonal loop, or vice versa: the quarter turn interchanges the two roles), so there are  $n(n-2)/8$  sub-classes.

sub-classes mod  $D_4$  at  $n = 23$ , 1956 and 78 at  $n = 27$ ): no configuration exists in them. We have no structural explanation.

## 4.2 New configurations with exactly half-turn symmetry

Table 3 lists the new configurations; their coordinates and their encodings in the 90-character format of [3] are given in the Appendix. “First located” refers to the public files of the database on 2026-08-11 (the database’s history page mentions further contributions in preparation at that time), and to a search of the literature; it awaits the maintainer’s confirmation. For  $n = 36$  both database cells were empty; for  $n = 37$  and 39 the column c (rct4) was populated, so the configurations below are new representatives of exact half-turn symmetry outside the rct4 structure, not the first configurations with that stabilizer; the  $n = 33$  configuration is new but the cell was not empty. All nine configurations were checked to be pairwise inequivalent (canonical forms under  $D_4$ ). The second and third  $n = 39$  configurations have their defects through the central class — they were found only after the correction of the generator described above.

| $n$ | family and defect                                                   | remark                                           |
|-----|---------------------------------------------------------------------|--------------------------------------------------|
| 36  | two loops: (6, 6), (29, 29) and (5, 30), (30, 5)                    | first located exact half-turn (both cells empty) |
| 36  | two loops: (1, 1), (34, 34) and (9, 26), (26, 9)                    | second, inequivalent                             |
| 36  | two loops: (1, 1), (34, 34) and (7, 28), (28, 7)                    | third, inequivalent                              |
| 37  | 3-cycle (1, 3), (3, 31), (31, 1), C4 base                           | first located outside rct4 (column :)            |
| 37  | 3-cycle (2, 4), (4, 20), (20, 34), C4 base                          | second, inequivalent                             |
| 39  | 3-cycle (4, 8), (8, 20), (20, 34), C4 base                          | first located outside rct4 (column :)            |
| 39  | 3-cycle (1, 5), (5, 19), (19, 1) through the central class, C4 base | second, inequivalent                             |
| 39  | 3-cycle (4, 7), (7, 19), (19, 4) through the central class, C4 base | third, inequivalent                              |
| 33  | 3-cycle (2, 3), (3, 19), (19, 2), C4 base                           | new (cell not empty)                             |

Table 3: New  $2n$ -point configurations with stabilizer exactly  $\{1, \rho^2\}$ .

## 4.3 Cost of finding one configuration, and growth

For orientation we report measured costs of finding one  $2n$ -point configuration on the  $n \times n$  grid with the solver of Section 3, on two consumer machines (Apple M3 laptop, 8 cores; a 12-core x86 server; 8–12 seeds in parallel; wall time to the first configuration): rot4 (even  $n$ ):  $n = 36$ : 10–60 s,  $n = 40$ : 70 s; rct4 (odd  $n$ ):  $n = 41$ : 806 s,  $n = 43$ : about 6.7 core-hours,  $n = 45$ : none within 3.3 core-hours; rot2:  $n = 24$ : 10 s,  $n = 27$ : 137 s. These are measured ratios for named instances, not a complexity law: the cost grows by roughly a factor 8–10 per +4 in  $n$  for rot4/rct4 and

by a factor of about 6 per +2 for rot2, consistent with the exhaustive tree sizes (rot4:  $\times 20$  per +4; rot2 and dia1:  $\times 9\text{--}10$  per +2) divided by the slowly growing numbers of configurations, and consistent with the growth reported for CP-SAT in [4]. Extrapolated — with the obvious uncertainty of an extrapolation over more than 25 steps — to the open sizes 71, 73, 75, 77, ... this suggests  $10^5\text{--}10^7$  core-hours for our program; the record computations of 2026 (SAT and GPU search on clusters) are of that order or use better tools. Sub-class sweeps of the defect families are cheap because a fixed defect shades a large part of the grid: a sub-class of the 3-cycle family at  $n = 37\text{--}41$  is exhausted in seconds to a few minutes.

#### 4.4 Negative results, with their scope

- *Consequence of the lemma.* A single defect pair off the diagonals is impossible (Corollary 2); the “rot4 except one pair anywhere” family is empty, as our enumerations for  $n \leq 27$  had shown before the lemma was found.
- *Computer-assisted, exhaustive.* The mixed families are empty for  $9 \leq n \leq 27$  (above); the two-loop family is empty for  $14 \leq n \leq 22$ ; the 3-cycle family is empty for  $n = 15, 19, 23, 25$ .
- *Bounded search, no conclusion about existence.* Shifting a known rot4 configuration for  $n = 44$  into the  $46 \times 46$  grid (which preserves all its collinearities) and freeing  $k$  of its 22 orbits, no completion exists for  $k \leq 15$  (exhaustive for those  $k$ ); local search on exactly  $2n$  points (min-conflicts with tabu, simulated annealing) failed from  $n \approx 12\text{--}28$  on with our implementations and budgets, although it produces large partial configurations easily (e.g. 134 points on the  $71 \times 71$  grid); belief-propagation decimation without backtracking failed from  $n = 8$  on. These are statements about the specified methods, not about the configurations.
- *Empirical observation.* For the rot4/rct4 configurations of the database with  $41 \leq n \leq 76$  (142 configurations) the largest subset lying on one line, axis-parallel parabola or shifted hyperbola over  $\mathbb{F}_p$ , maximised over primes  $11 \leq p \leq 61$ , is on average 16% of the points (never above 22%), *less* than for random sets with two points per row (19% on average): relative to these curve families and primes, the finite extremal configurations do not reduce modulo a prime to a low-degree curve (cf. Green’s remark on Problem 72 of [11]), while the best asymptotic constructions do. On the algebraic side, the modular-hyperbola construction of [7] is optimal within its own  $2p \times 2p$  box, unions of two hyperbolae gain only a bounded amount as far as computed, and for  $H(1) \cup H(-1)$  a lawful subset has at most  $(11/3 + o(1))(p - 1)$  points (an explicit line cover); see the companion note [14].

## 5 Reproducibility

The programs, the sub-class lists, the sweep journals and the certified configurations are in the public repository <https://github.com/iwasborninbali/saturation> (immutable tag `defects-v0.7`; checksums in `docs/RELEASES.md`):

`there.c`; `kit71/there_twisted_experiment.c` and its even- $n$  variant `bench/there_tw_even.c` (the two bases and the PAIRS defect built in; validated against full enumerations of the rot4 and dia2 classes); the generators `bench/defects3.py` (all 3-cycle defects mod  $D_4$ , central class included) and `bench/mixed3.py`; the sweep drivers `bench/family_sweep2.sh`, `central_sweep.sh`, `mixed_sweep.sh`, `twoloop_exh.sh` (same directory); the finished journals under `logs/sweeps/` (one line per sub-class with the solver’s termination status; BOOK lines with every labelled solution); `web/stab.py` for the exact class and the database encoding; `docs/SUBMISSION.md` for the configurations; `make check` for a one-command test. Typical commands: `ALL=1 PAIRS="1,3;3,31;31,1" ./there 37 2` enumerates the sub-class that contains the first  $n = 37$  configuration (one configuration,  $1.4 \cdot 10^7$  nodes); `PAIRS="6,6;5,30" ./there 36 2` finds the

first  $n = 36$  configuration in about 100s and `PAIRS="1,1;9,26" ./there 36 2` the second in about 30s (wall time on the laptop); the sub-class  $\{5, 6\}$  is exhausted in about 7 min and contains exactly two labelled configurations, i.e. one up to the diagonal reflections, which preserve both loops. Journals of the early sweeps run on the first agent’s machine (the first halves of the lists for  $n = 39, 41, 45$ ) are being added; the  $n = 33$  row was rerun in full with the corrected generator on a cloud machine (`logs/sweeps/family3full.*n33.txt`: 2360 sub-classes per base, one configuration), and the  $n = 37$  row was completed by sweeping the 153 central sub-classes for both bases. All exhaustive statements are computer-assisted results of a trusted program (validated as in Section 3); no independent certificates of non-existence are produced.

## 6 AI/tool-use disclosure and authorship statement

Following the arXiv policy on generative AI, the AI systems involved are not listed as authors; their role is disclosed here. This work was carried out by autonomous AI agents (Claude Fable 5, Anthropic), running as independent solvers on separate machines and communicating through the repository, under the direction of the author of record; a third agent of the same kind worked mainly on the companion note [14]. The agents wrote the programs, designed and ran the searches, found and proved the balance lemma (independently of each other, in two formulations), verified each other’s results, and drafted and revised this text, including the corrections after two external LLM-assisted audits commissioned by the author of record; the author of record posed the problem, provided the computing environment, made all decisions about scope and publication, and takes responsibility for the content; the extent of his independent verification of the proofs and of the computations is recorded in `docs/HUMAN_AUDIT.md` of the repository and will be stated precisely in any submitted version. We estimate that about nine tenths of the technical work was done by the AI systems; we state this plainly because it is true. All configurations reported here are verifiable by anyone with the scripts in the repository (every triple checked in exact integer arithmetic).

## Acknowledgements

We thank Achim Flammenkamp for maintaining the no-three-in-line database and its class counts, without which the empty cells would not have been visible, and Thomas Prellberg for the detailed account of the CP-SAT computations.

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## A The configurations

Rows and columns are numbered  $0, \dots, n - 1$  from the top-left corner; the encoding is the class character followed by two column characters per row, with the 90-character alphabet of [3]:

0-9A-Za-z#%&@?!()[]<>{ }=+|-/~^\_~.;,.,.

$n = 36$  (**configuration 1**), **72 points, stabilizer**  $\{1, \rho^2\}$ . Encoding:  
:3NEHFSVZ3FPU6E2BNQ8I5GDS08JOTYVXDP9Y1QAM2416BGRZ7MJUHR9COXLT5AKW047KILCW

Rows  $u$  :  $v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 3,23   1: 14,17   2: 15,28   3: 31,35   4: 3,15   5: 25,30   6: 6,14   7: 2,11  
8: 23,26   9: 8,18   10: 5,16   11: 13,28   12: 0,8   13: 19,24   14: 29,34   15: 31,33  
16: 13,25   17: 9,34   18: 1,26   19: 10,22   20: 2,4   21: 1,6   22: 11,16   23: 27,35  
24: 7,22   25: 19,30   26: 17,27   27: 9,12   28: 24,33   29: 21,29   30: 5,10   31: 20,32  
32: 0,4   33: 7,20   34: 18,21   35: 12,32

$n = 36$  (**configuration 2**), **72 points, stabilizer**  $\{1, \rho^2\}$ . Encoding:  
:JK1J9OCK6AORRVEH56QXLV25MWCIAS0301DS7MYZWZ7PHN3DUX4E29TUIL488BPTFNBQGYFG

Rows  $u$  :  $v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 19,20   1: 1,19   2: 9,24   3: 12,20   4: 6,10   5: 24,27   6: 27,31   7: 14,17  
8: 5,6   9: 26,33   10: 21,31   11: 2,5   12: 22,32   13: 12,18   14: 10,28   15: 0,3  
16: 0,1   17: 13,28   18: 7,22   19: 34,35   20: 32,35   21: 7,25   22: 17,23   23: 3,13  
24: 30,33   25: 4,14   26: 2,9   27: 29,30   28: 18,21   29: 4,8   30: 8,11   31: 25,29  
32: 15,23   33: 11,26   34: 16,34   35: 15,16

$n = 36$  (configuration 3), 72 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:DK1GGNLRBK7RNPUS35AH6QIV26LZ3D04XYBQ9012VZMW0ETX4H9TIPUW57AC8SF08ECJJYFM

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 13,20 1: 1,16 2: 16,23 3: 21,27 4: 11,20 5: 7,27 6: 23,25 7: 28,30  
8: 3,5 9: 10,17 10: 6,26 11: 18,31 12: 2,6 13: 21,35 14: 3,13 15: 0,4  
16: 33,34 17: 11,26 18: 9,24 19: 1,2 20: 31,35 21: 22,32 22: 0,14 23: 29,33  
24: 4,17 25: 9,29 26: 18,25 27: 30,32 28: 5,7 29: 10,12 30: 8,28 31: 15,24  
32: 8,14 33: 12,19 34: 19,34 35: 15,22

$n = 37$  (configuration 1), 74 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:GJ39EP7VENDZJRSX7P6ZCF28IQ4VWYKQFa06COUa0LAG245WAISYL01UBT389H1NDM5TBMRXHK

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 16,19 1: 3,9 2: 14,25 3: 7,31 4: 14,23 5: 13,35 6: 19,27 7: 28,33  
8: 7,25 9: 6,35 10: 12,15 11: 2,8 12: 18,26 13: 4,31 14: 32,34 15: 20,26  
16: 15,36 17: 0,6 18: 12,24 19: 30,36 20: 0,21 21: 10,16 22: 2,4 23: 5,32  
24: 10,18 25: 28,34 26: 21,24 27: 1,30 28: 11,29 29: 3,8 30: 9,17 31: 1,23  
32: 13,22 33: 5,29 34: 11,22 35: 27,33 36: 17,20

$n = 37$  (configuration 2), 74 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:FJ7E4PBGK06U5VDZDH9RCF2X4QSTIZQa2X0SEM8a3Y0A1I78AW3YL09RJN1N5V6UCGKPBWMTHL

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 15,19 1: 7,14 2: 4,25 3: 11,16 4: 20,24 5: 6,30 6: 5,31 7: 13,35  
8: 13,17 9: 9,27 10: 12,15 11: 2,33 12: 4,26 13: 28,29 14: 18,35 15: 26,36  
16: 2,33 17: 0,28 18: 14,22 19: 8,36 20: 3,34 21: 0,10 22: 1,18 23: 7,8  
24: 10,32 25: 3,34 26: 21,24 27: 9,27 28: 19,23 29: 1,23 30: 5,31 31: 6,30  
32: 12,16 33: 20,25 34: 11,32 35: 22,29 36: 17,21

$n = 39$  (configuration 1), 78 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:CGAWLU7M8CAN1FPZ2K9TXbOPYc7BBL5W3c2E4JIKJY0a0Z6XHRRV04DE159T1a3DNbFSQUGV8H6SMQ

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 12,16 1: 10,32 2: 21,30 3: 7,22 4: 8,12 5: 10,23 6: 1,15 7: 25,35  
8: 2,20 9: 9,29 10: 33,37 11: 24,25 12: 34,38 13: 7,11 14: 11,21 15: 5,32  
16: 3,38 17: 2,14 18: 4,19 19: 18,20 20: 19,34 21: 24,36 22: 0,35 23: 6,33  
24: 17,27 25: 27,31 26: 0,4 27: 13,14 28: 1,5 29: 9,29 30: 18,36 31: 3,13  
32: 23,37 33: 15,28 34: 26,30 35: 16,31 36: 8,17 37: 6,28 38: 22,26

$n = 39$  (configuration 2), 78 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:BI5HCWEQST8J2MSU7X4M47Dc3aKRNZEh69NbDc1bOP1FTWL03FBI2Z0PVYGY5V8AGaJU9AC06QLXKR

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 11,18 1: 5,17 2: 12,32 3: 14,26 4: 28,29 5: 8,19 6: 2,22 7: 28,30  
8: 7,33 9: 4,22 10: 4,7 11: 13,38 12: 3,36 13: 20,27 14: 23,35 15: 14,17  
16: 6,9 17: 23,37 18: 13,38 19: 1,37 20: 0,25 21: 1,15 22: 29,32 23: 21,24  
24: 3,15 25: 11,18 26: 2,35 27: 0,25 28: 31,34 29: 16,34 30: 5,31 31: 8,10  
32: 16,36 33: 19,30 34: 9,10 35: 12,24 36: 6,26 37: 21,33 38: 20,27

$n = 39$  (configuration 3), 78 points, stabilizer  $\{1, \rho^2\}$ . Encoding:  
:AEIUHR9P678GRYJP1XHZNc26CQ37NcAEIXTaMb4Y1G295KOS0FVZCQWaf3L5bDJ4BMUVWDTBL8KOS

Rows  $u : v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 10,14 1: 18,30 2: 17,27 3: 9,25 4: 6,7 5: 8,16 6: 27,34 7: 19,25  
8: 1,33 9: 17,35 10: 23,38 11: 2,6 12: 12,26 13: 3,7 14: 23,38 15: 10,14  
16: 18,33 17: 29,36 18: 22,37 19: 4,34 20: 1,16 21: 2,9 22: 5,20 23: 24,28

24: 0,15   25: 31,35   26: 12,26   27: 32,36   28: 0,15   29: 3,21   30: 5,37   31: 13,19  
 32: 4,11   33: 22,30   34: 31,32   35: 13,29   36: 11,21   37: 8,20   38: 24,28

$n = 33$ , **66 points, stabilizer**  $\{1, \rho^2\}$ . Encoding:

:MOAC3H9J6GJOB9B05PT0VPQH5UEI2C4SKUEI2R1F671W37RWLN4L8DGQDNFTKM8A

Rows  $u$  :  $v_1, v_2$  (row  $u$  has points at columns  $v_1, v_2$ ):

0: 22,24   1: 10,12   2: 3,17   3: 9,19   4: 6,16   5: 19,24   6: 11,28   7: 9,11  
 8: 0,5   9: 25,29   10: 0,31   11: 25,26   12: 17,31   13: 5,30   14: 14,18   15: 2,12  
 16: 4,28   17: 20,30   18: 14,18   19: 2,27   20: 1,15   21: 6,7   22: 1,32   23: 3,7  
 24: 27,32   25: 21,23   26: 4,21   27: 8,13   28: 16,26   29: 13,23   30: 15,29   31: 20,22  
 32: 8,10